

Computing the SVD by Hand

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1. Overview

Key point: Multiplying A by its own transpose makes one of the two orthogonal matrices disappear. The eigenvectors of $A^T A$ are the right singular vectors, its eigenvalues are the squared singular values, and $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ gives the left singular vectors.

This Note follows MIT OpenCourseWare 18.06 Linear Algebra, Lecture 29 (Gilbert Strang), and *Mathematics for Machine Learning* (Deisenroth, Faisal, Ong, 2020), §4.5.2.

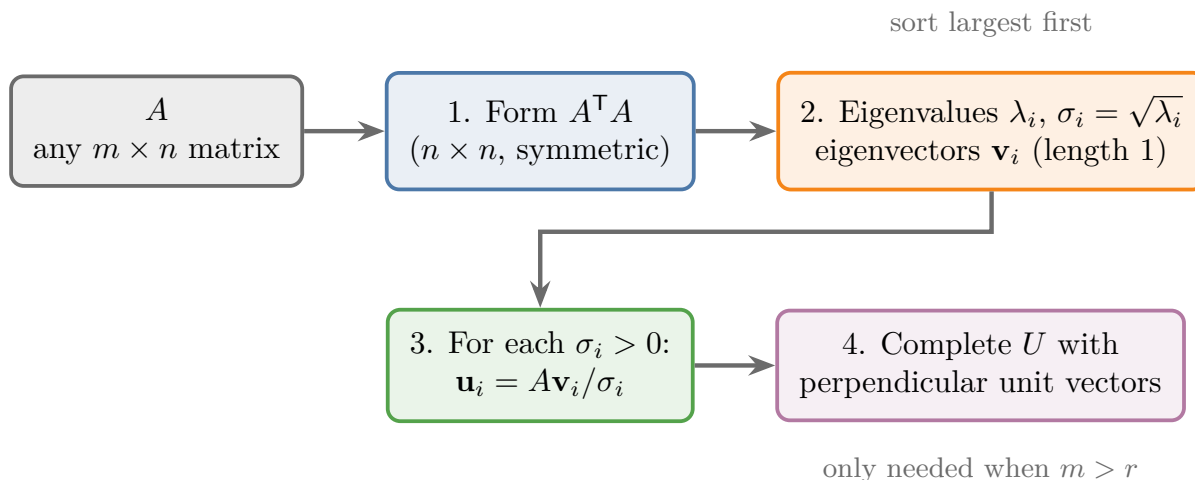


Figure 1: The four steps of computing an SVD by hand

The first SVD Note, the [SVD geometry Note](#) showed what $A = U\Sigma V^T$ means: rotate, stretch, rotate. It gave the factors of

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$

without saying where they came from. This Note finds them:

- why $A^T A$ and AA^T hold the answer (Section 2);
- the four-step recipe of Figure 1, worked on A (Section 3);
- a sign trap, and how $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ avoids it (Section 4);
- a matrix of rank 1, where a singular value is 0 (Section 5);
- the four subspaces that the SVD gives bases for (Section 6);
- a matrix that is not square (Section 7);
- why computers do not use this recipe (Section 8).

Finding eigenvalues and eigenvectors by hand, with $\det(A - \lambda I) = 0$, is taught in the [eigenvectors and eigenvalues Note](#); we use it here without repeating it.

2. Making one orthogonal matrix disappear

Key point: In $A^T A$ the two copies of U meet as $U^T U = I$ and vanish, leaving an eigen-decomposition of a symmetric matrix.

2.1 $A^T A$ gives V and the singular values

Key point: $A^T A = V \Sigma^T \Sigma V^T$: its eigenvectors are the $\mathbf{v}_i$, its eigenvalues are the σ_i^2 .

$A = U \Sigma V^T$ has two unknown orthogonal matrices. Finding both at once is hard, so we look for an expression in which one of them disappears.

1. **In words:** transpose A (the transpose of a product is the product of the transposes in reverse order), multiply by A , and use $U^T U = I$ (see the [SVD geometry Note](#), section 3.3).
2. **Formula:**

$$A^T A = (V \Sigma^T U^T)(U \Sigma V^T) = V \Sigma^T \Sigma V^T = V \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2 \end{bmatrix} V^T$$

3. **Example:** for A with rows $[3, 0]$ and $[4, 5]$,

$$A^T A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$$

The right side of the formula is exactly an eigen-decomposition PDP^{-1} (see the [eigenvectors and eigenvalues Note](#), section 6.2), with $P = V$ and $P^{-1} = V^T$. So:

- the right singular vectors $\mathbf{v}_i$ are the eigenvectors of $A^T A$;
- the squared singular values σ_i^2 are its eigenvalues, so $\sigma_i = \sqrt{\lambda_i}$.

2.2 Why this always works

Key point: $A^T A$ is symmetric with eigenvalues ≥ 0 , so it always has orthonormal eigenvectors and real square roots.

A itself may have no useful eigenvectors, but $A^T A$ always does, for two reasons.

- **It is symmetric.** Its transpose is $A^T(A^T)^T = A^T A$, itself. A symmetric matrix always has a full set of perpendicular eigenvectors (the same fact that makes PCA work on a covariance matrix, see the [PCA step by step Note](#), section 4.3). So V can always be built as an orthogonal matrix.
- **Its eigenvalues are never negative.** For a unit eigenvector $\mathbf{v}$ with eigenvalue λ , $\lambda = \mathbf{v}^T A^T A \mathbf{v} = \|A \mathbf{v}\|^2 \geq 0$. So the square root $\sigma = \sqrt{\lambda}$ is a real, non-negative number. The formula also says σ is the length of $A \mathbf{v}$, as in the geometry Note.

So $A^T A$ is symmetric and positive semi-definite, for every matrix A of any shape.

2.3 AA^T gives U , with the same numbers

Key point: Multiplying in the other order removes V instead: $AA^T = U\Sigma\Sigma^T U^T$.

By the same steps, $AA^T = U\Sigma V^T V\Sigma^T U^T = U\Sigma\Sigma^T U^T$. So the left singular vectors $\mathbf{u}_i$ are the eigenvectors of AA^T , and its non-zero eigenvalues are the same σ_i^2 .

For A : AA^T has rows $[9, 12]$ and $[12, 41]$, a different matrix from $A^T A$. Yet both have the eigenvalues 45 and 5. This is no accident: AB and BA always share their non-zero eigenvalues.

3. The recipe, worked on a 2×2 matrix

Key point: Eigen-decompose $A^T A$, take square roots, then get each $\mathbf{u}_i$ by applying A to $\mathbf{v}_i$ and dividing by σ_i .

Figure 1 shows the four steps. We follow them for A with rows $[3, 0]$ and $[4, 5]$.

Step 1: form $A^T A$. As in Section 2.1, it has rows $[25, 20]$ and $[20, 25]$.

Step 2: eigenvalues and eigenvectors of $A^T A$.

1. **In words:** subtract λ from the diagonal and set the determinant to zero; then find the line of vectors each eigenvalue squishes to zero, and scale to length 1.

2. **Formula:**

$$\det(A^T A - \lambda I) = 0, \quad \sigma_i = \sqrt{\lambda_i}$$

3. **Example:**

$$\det \begin{bmatrix} 25 - \lambda & 20 \\ 20 & 25 - \lambda \end{bmatrix} = (25 - \lambda)^2 - 400 = 0 \implies 25 - \lambda = \pm 20$$

so $\lambda_1 = 45$ and $\lambda_2 = 5$ (largest first), and

$$\sigma_1 = \sqrt{45} = 3\sqrt{5} \approx 6.708, \quad \sigma_2 = \sqrt{5} \approx 2.236$$

For $\lambda_1 = 45$, $A^T A - 45I$ has rows $[-20, 20]$ and $[20, -20]$, which sends $[x, y]$ to zero when $x = y$. For $\lambda_2 = 5$, $A^T A - 5I$ has rows $[20, 20]$ and $[20, 20]$, which needs $y = -x$. Scaled to length 1:

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

They are perpendicular, as Section 2.2 promised.

Step 3: $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$. This is the singular value equation $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$ solved for $\mathbf{u}_i$.

1. **In words:** apply A to each right singular vector and divide by its singular value. The result is automatically a unit vector.

2. Formula:

$$\mathbf{u}_i = \frac{1}{\sigma_i} A \mathbf{v}_i$$

3. Example:

$$\mathbf{u}_1 = \frac{1}{3\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

The $\mathbf{u}$'s come out perpendicular without any extra work. In symbols: $(A\mathbf{v}_1)^\top(A\mathbf{v}_2) = \mathbf{v}_1^\top(A^\top A \mathbf{v}_2) = \lambda_2 \mathbf{v}_1^\top \mathbf{v}_2 = 0$.

Step 4: complete U . Here there are already two $\mathbf{u}$'s for a 2×2 matrix, so nothing is missing. Sections 5 and 7 need this step.

Check. Multiplying the factors back together:

$$U\Sigma = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -3 \\ 9 & 1 \end{bmatrix}$$

$$U\Sigma V^\top = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -3 \\ 9 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 6 & 0 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = A$$

Python: Checking each step with NumPy.

```
import numpy as np
```

```
A = np.array([[3, 0], [4, 5]])
lam, V = np.linalg.eigh(A.T @ A)      # eigh: for symmetric
lam, V = lam[:, : -1], V[:, : -1]    # largest first
sigma = np.sqrt(lam)                 # [6.708, 2.236]
U = A @ V / sigma                    # column i = A v_i / sigma_i
U @ np.diag(sigma) @ V.T             # gives back A
A @ V / sigma divides column i of AV by  $\sigma_i$ : NumPy matches the 1D array sigma with
the columns.
```

4. The sign trap

Key point: Finding U and V separately as eigenvectors can pick mismatched signs and give a wrong product; computing $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ always matches them.

Section 2.3 suggests a shortcut: take the $\mathbf{u}$'s as the eigenvectors of $AA^\top$, found on their own. For A , $AA^\top$ has rows $[9, 12]$ and $[12, 41]$, with eigenvalues 45 and 5. Its eigenvectors are the lines through $[1, 3]$ and through $[3, -1]$.

Every non-zero multiple of an eigenvector is an eigenvector too, so $\frac{1}{\sqrt{10}}[3, -1]$ is as correct an eigenvector as $\frac{1}{\sqrt{10}}[-3, 1]$. But it is the negative of the $\mathbf{u}_2$ that belongs with our $\mathbf{v}_2$. Using it gives

$$\frac{1}{\sqrt{10}} \begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ 5 & 4 \end{bmatrix} \neq A$$

Each piece is a valid eigenvector, yet the product is a different matrix. The pairs $(\mathbf{u}_i, \mathbf{v}_i)$ must match: flipping both signs is allowed, flipping one is not (see the [SVD geometry Note](#), section 5.1). The eigenvalue problem for $AA^\top$ knows nothing about V , so it cannot keep the pairs matched.

The fix is Step 3: compute $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$ from the $\mathbf{v}_i$ already chosen. Then $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$ holds by construction, whatever sign each $\mathbf{v}_i$ was given.

5. A matrix of rank 1

Key point: A rank-1 matrix has one non-zero singular value; the zero one has no $\mathbf{u}$ from $A\mathbf{v}/\sigma$, so we complete U with a perpendicular unit vector.

Take

$$C = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

The second row is twice the first, so the rank is 1: C squishes the whole plane onto one line, the line through $[1, 2]$ (its columns $[2, 4]$ and $[1, 2]$ both lie on it).

Step 1.

$$C^\top C = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix}$$

Step 2. Its determinant is $20 \times 5 - 10 \times 10 = 0$, so one eigenvalue is 0. The eigenvalues of a 2×2 matrix add up to its diagonal sum, $20 + 5 = 25$, so the other is 25. So $\sigma_1 = 5$ and $\sigma_2 = 0$, and the eigenvectors are

$$\mathbf{v}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$\mathbf{v}_2$ is the direction C squishes to nothing: $C[-1, 2] = [-2 + 2, -4 + 4] = [0, 0]$.

Step 3. Only σ_1 is non-zero:

$$\mathbf{u}_1 = \frac{1}{5} C \mathbf{v}_1 = \frac{1}{5\sqrt{5}} \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Step 4. $\mathbf{u}_2 = C\mathbf{v}_2/\sigma_2$ would divide zero by zero. Any unit vector perpendicular to $\mathbf{u}_1$ completes U to an orthogonal matrix; we take $\mathbf{u}_2 = \frac{1}{\sqrt{5}}[-2, 1]$. It is multiplied by $\sigma_2 = 0$, so it never affects the product.

Result.

$$C = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$$

Only the first column of U and the first row of V^T meet a non-zero number, so the product shrinks to

$$C = 5 \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

A column times a row is a whole matrix of rank 1. The [low-rank approximation Note](#) builds every matrix out of such pieces.

Python: NumPy reports the zero singular value.

```
C = np.array([[2, 1], [4, 2]])
```

```
np.linalg.svd(C)[1]          # [5., 0.]
```

```
np.linalg.matrix_rank(C)    # 1
```

`matrix_rank` from the [linear combinations, span and basis Note](#) works exactly this way: it counts the singular values that are not (almost) zero.

6. The four subspaces

Key point: The $\mathbf{v}$'s split the input space into the row space and the null space; the $\mathbf{u}$'s split the output space into the column space and the left null space. The SVD gives perpendicular bases for all four.

The rank-1 example shows a pattern that holds for every matrix. Figure 2 draws it for C .

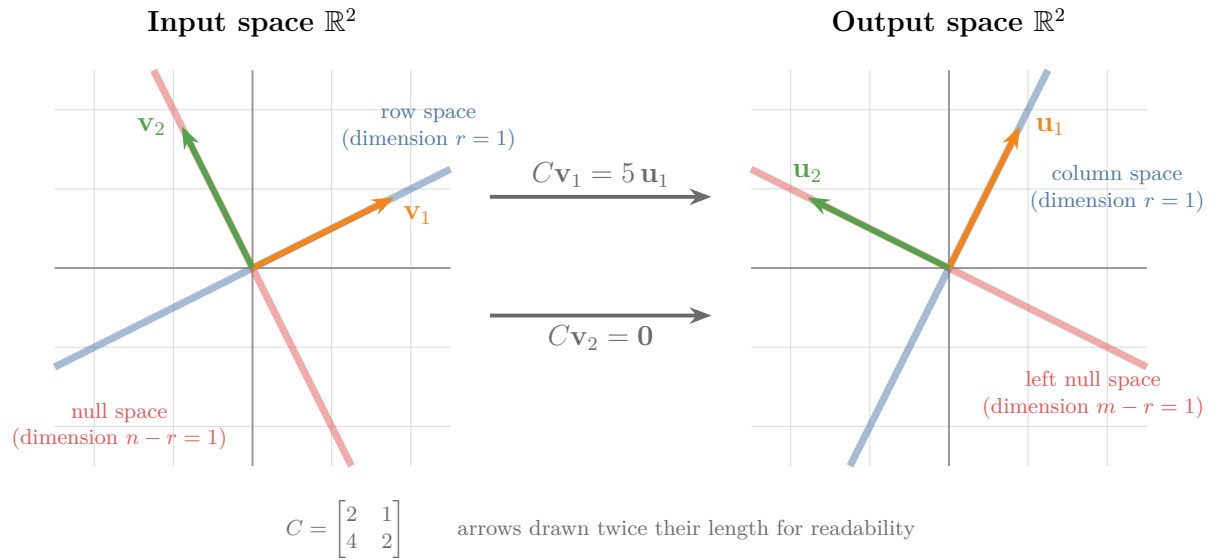


Figure 2: The SVD of C gives a basis for each of its four subspaces

For an $m \times n$ matrix of rank r (there are r non-zero singular values):

Subspace	Where	Basis from the SVD	Dimension	For C
Row space (span of the rows)	input $\mathbb{R}^n$	$\mathbf{v}_1, \dots, \mathbf{v}_r$	r	line through $[2, 1]$
Null space (vectors sent to $\mathbf{0}$)	input $\mathbb{R}^n$	$\mathbf{v}_{r+1}, \dots, \mathbf{v}_n$	$n - r$	line through $[-1, 2]$
Column space (span of the columns)	output $\mathbb{R}^m$	$\mathbf{u}_1, \dots, \mathbf{u}_r$	r	line through $[1, 2]$
Left null space (outputs never reached, perpendicular to them)	output $\mathbb{R}^m$	$\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$	$m - r$	line through $[-2, 1]$

The column space is the span of the columns from the [linear transformations and matrices Note](#) (section 6): every output lands in it. The null space is everything the matrix squishes to the origin.

What makes the SVD's bases special is that they line up in pairs: A sends $\mathbf{v}_1$ to a multiple of $\mathbf{u}_1$, $\mathbf{v}_2$ to a multiple of $\mathbf{u}_2$, and so on, with no mixing. The null-space $\mathbf{v}$'s go to zero, which is

where the zeros on the diagonal of Σ come from. Any method of making perpendicular bases could give bases of the four subspaces; only the SVD's bases also make the matrix diagonal.

7. A matrix that is not square

Key point: For a tall matrix, $A^T A$ is small; the extra columns of U come from completing it with perpendicular unit vectors.

Take the 3×2 matrix B of the [SVD geometry Note](#) (section 6):

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Steps 1 and 2. $B^T B$ is only 2×2 :

$$B^T B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad (2 - \lambda)^2 - 1 = 0 \implies \lambda = 3, 1$$

So $\sigma_1 = \sqrt{3} \approx 1.732$ and $\sigma_2 = 1$, with $\mathbf{v}_1 = \frac{1}{\sqrt{2}}[1, 1]$ and $\mathbf{v}_2 = \frac{1}{\sqrt{2}}[1, -1]$.

Step 3.

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}} B \mathbf{v}_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 0.816 \\ 0.408 \\ 0.408 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{1} B \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

Step 4. U is 3×3 and needs a third column, perpendicular to both. $\mathbf{u}_3 = \frac{1}{\sqrt{3}}[1, -1, -1]$ works: its dot products with $[2, 1, 1]$ and with $[0, -1, 1]$ are both 0. It spans the left null space, the one direction in 3D that B never reaches.

The other route, eigenvectors of BB^T , would mean a 3×3 eigenvalue problem (eigenvalues 3, 1 and 0), and would still risk the sign trap of Section 4. For a tall data matrix with thousands of rows, $A^T A$ is the small one, so we always start from it.

8. How computers compute the SVD

Key point: Forming $A^T A$ squares the singular values, and small ones get lost in rounding. Libraries compute the SVD directly from A .

The recipe is for understanding and for small examples. On a computer it has a flaw. Numbers are stored with about 16 significant digits. Squaring a singular value of 10^{-9} gives 10^{-18} ; next to an eigenvalue of 4, that is far below the 16th digit and disappears in rounding.

Python: A nearly rank-1 matrix: its second singular value is tiny but not zero.

```
M = np.array([[1, 1], [1, 1 + 1e-9]])
np.linalg.svd(M)[1]           # [2.0, 5.0e-10]: correct
np.sqrt(np.linalg.eigvalsh(M.T @ M))
# [0.0, 2.0]: the small one is lost
```

So `np.linalg.svd` never forms $A^T A$. It reduces A itself step by step with orthogonal matrices, which do not magnify rounding errors, until the singular values can be read off.

Extra: The ratio σ_1/σ_n of the largest to the smallest singular value is the **condition number** of a matrix. It measures how much errors in the input can be magnified by solving with that matrix. Forming $A^T A$ squares it: a condition number of 10^8 becomes 10^{16} , which uses up all 16 digits. This is why the least-squares solvers of the [multiple linear regression code Note](#) (section 6) prefer to work from X itself rather than from $X^T X$. `np.linalg.cond(A)` computes it.

9. Summary

Step	What we compute	For A (rows $[3, 0]$, $[4, 5]$)	For C (rows $[2, 1]$, $[4, 2]$)
1	$A^T A$	rows $[25, 20]$, $[20, 25]$	rows $[20, 10]$, $[10, 5]$
2	eigenvalues $\rightarrow \sigma_i = \sqrt{\lambda_i}$	$45, 5 \rightarrow 6.708, 2.236$	$25, 0 \rightarrow 5, 0$
2	eigenvectors $\mathbf{v}_i$	$\frac{1}{\sqrt{2}}[1, 1]$, $\frac{1}{\sqrt{2}}[-1, 1]$	$\frac{1}{\sqrt{5}}[2, 1]$, $\frac{1}{\sqrt{5}}[-1, 2]$
3	$\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$	$\frac{1}{\sqrt{10}}[1, 3]$, $\frac{1}{\sqrt{10}}[-3, 1]$	$\frac{1}{\sqrt{5}}[1, 2]$
4	complete U	nothing missing	$\frac{1}{\sqrt{5}}[-2, 1]$

- $A^T A = V\Sigma^T\Sigma V^T$: eigenvectors are the $\mathbf{v}_i$, eigenvalues the σ_i^2 .
- $A^T A$ is always symmetric and positive semi-definite, so this always works.
- Get the $\mathbf{u}_i$ from $A\mathbf{v}_i/\sigma_i$, not from a separate eigen-problem: that keeps the signs matched.
- Zero singular values belong to the null space; the missing $\mathbf{u}$'s are any perpendicular completion.
- The SVD gives perpendicular bases for the row space, null space, column space and left null space.
- Computers compute the SVD from A directly, because $A^T A$ loses small singular values.

10. Key terms

Term	Meaning
$A^T A$	The symmetric, positive semi-definite matrix whose eigenvectors are the right singular vectors and whose eigenvalues are the squared singular values
Row space	The span of the rows of a matrix; spanned by the $\mathbf{v}_i$ with $\sigma_i > 0$
Null space	All vectors a matrix sends to $\mathbf{0}$; spanned by the $\mathbf{v}_i$ with $\sigma_i = 0$
Column space	The span of the columns: every possible output; spanned by the $\mathbf{u}_i$ with $\sigma_i > 0$
Left null space	The output directions perpendicular to every column; spanned by the remaining $\mathbf{u}_i$
Four fundamental subspaces	Row space, null space, column space and left null space of a matrix
Condition number	σ_1/σ_n : how much a matrix can magnify errors when we solve with it